

ELEANOR BYLER

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EXPERIENCE

Senior Data Scientist and Team Lead

Pacific Northwest National Laboratory

📅 May 2020 – Present 📍 Seattle, WA

- Develop and deliver novel, state-of-the-art AI/ML solutions for diverse applications across the national security domain. Technical lead for four programs; responsible for tasking research team, delivering technical products, and briefing government sponsors.
- Expertise in overhead imagery and geospatial data, limited data (few-shot learning), and fusing data modalities (LIDAR, multi/hyperspectral imagery). Emphasis on computer vision applications: rare object detection and tracking, fine-grained classification, vision-based regression.

Postdoctoral Research Fellow

Australian National University, Research School for Astrophysics

📅 January 2018 – March 2020 📍 Canberra, Australia

- Designed innovative, flexible, star formation model to improve widely-used galaxy model (800+ citations; 🐍 python-fsps). Developed open-source python toolkit for processing simulations (🌥 cloudy-fsps).

PROJECTS

Rare Object Detection in Overhead Imagery

Pacific Northwest National Laboratory (sponsor-funded)

- Developed machine learning pipeline to ingest overhead imagery into a suite of custom deep learning models, optimized to scale for big data and deployed to sponsor production environments for active monitoring.
- Model designed to detect a singular rare object under tight false alarm rate constraints; leverages unique, identifiable features despite limited training data.

Remote Quantification of Hazardous Plumes

Pacific Northwest National Laboratory (sponsor-funded)

- Built and trained DL models to predict the quantity of released sulfur dioxide from volcanic plume imagery. Successfully transferred model from images of volcanic plumes to other hazardous releases (📄 paper).

Robustness of Multi-Input Deep Learning Models

Pacific Northwest National Laboratory

- Developed and evaluated deep data fusion architectures for optical imagery combined with LIDAR data or hyperspectral imagery and domain appropriate augmentations (📄 paper).
- Quantified the robustness of multi-input models against adversarial attacks (data poisoning, physically realistic perturbations) and environmental distribution shifts (🌥 gummy; 📄 arXiv:2309.12463, arXiv:2305.11347).

SKILLS

Deep Learning Geospatial Intelligence
Machine Learning Computer Vision
Remote Sensing Hyperspectral Data
Satellite Imagery Multi-modal learning
Communication Data Visualization

NumPy pandas PyTorch rasterio
gdal SciPy scikit-learn shapely
GeoPandas opencv TensorFlow
Git SQL Linux GIS Docker

EDUCATION

PhD in Astrophysics

University of Washington

📅 2017

EXTRAS

- 💰 **Selected Grants**
- > Federal Grant (\$750K)
 - > Lab-directed R&D Fund (\$500K)
 - > NASA ROSES (\$250 K)
 - > Hubble Space Telescope (\$169 K)

- 📄 **Research Impact**
- > 23 publications (4000+ citations).
 - > Leading division-level capability development for GEOINT programs.
 - > Deep learning SME and reviewer for internally funded R&D proposals.

- 🎤 **Communication**
- > Designed & led deep learning workshops internally & for gov sponsors.
 - > Co-founder of "Astronomy on Tap," Seattle's popular live science event.
 - > 6 invited talks and 15 contributed talks at professional conferences.
 - > Many public talks: Seattle MoPOP, Olympic National Park, Nerd Nite, DOE's National Science Bowl.

Active security clearance

Publications

23. **Byler, E.**, Forland, F., McKay, M. (2024). “Chemical signature characterization with hyperspectral imagery: novel deep learning model architectures and physically-motivated data augmentation techniques,” *SPIE Defense and Commercial Sensing*, 13031, 138–148. [DOI](#).
22. Svinth, C., Chojnicki, K., & **Byler, E.** (2024). “Deep learning for hazard quantification from imagery,” *SPIE Defense and Commercial Sensing*, 13056, 208–218. [DOI](#).
21. Kono, N., **Byler, E.**, et al. (2024). “Adapting self-supervised learning to the hyperspectral domain: methods, challenges, and lessons learned,” *SPIE Defense and Commercial Sensing*, 13031, 130310J. [DOI](#).
20. Engel, A., Narayan, G., & **Byler, N.** (2024). “Preliminary Report on Mantis Shrimp: a Multi-Survey Computer Vision Photometric Redshift Model,” *arXiv*, arXiv:2402.03535. [arXiv](#).
19. Godfrey, C., ... & **Byler, E.** (2023). “Impact of model architecture on robustness and interpretability of multispectral deep learning models,” *SPIE Defense and Commercial Sensing*, 12519, 184–199. [arXiv](#).
18. Bishoff, E., ... & **Byler, E.** (2023). “Quantifying the robustness of deep multispectral segmentation models against natural perturbations and data poisoning,” *SPIE Defense and Commercial Sensing*, 12519, 200–213. [arXiv](#).
17. Godfrey, C., ... & **Byler, E.** (2023). “How many dimensions are required to find an adversarial example?,” *CVPR Workshop Proceedings*, 2352–2359. [arXiv](#).
16. Garg, P., et al. including **N. Byler** (2022). “The BPT Diagram in Cosmological Galaxy Formation Simulations: Understanding the Physics Driving Offsets at High Redshift,” *ApJ*, 926, 80. [ADS](#).
15. Johnson, B., et al. including **N. Byler** (2021). “dfm/python-fsps: python-fsps v0.4.1rc1,” *zndo*,. [ADS](#).
14. **Byler, N.**, Kewley, L. J., Rigby, J. R., Acharyya, A., Berg, D. A., Bayliss, M., & Sharon, K. (2020). “A Comparison of UV and Optical Metallicities in Star-forming Galaxies,” *ApJ*, 893, 1. [ADS](#).
13. McLeod, A. F., et al. including **N. Byler** (2020). “Stellar Feedback and Resolved Stellar IFU Spectroscopy in the Nearby Spiral Galaxy NGC 300,” *ApJ*, 891, 25. [ADS](#).
12. **Byler, N.**, Dalcanton, J. J., Conroy, C., Johnson, B. D., Choi, J., Dotter, A., & Rosenfield, P. (2019). “Self-consistent Predictions for LIER-like Emission Lines from Post-AGB Stars,” *AJ*, 158, 2. [ADS](#).
11. D’Agostino, J. J., Kewley, L. J., Groves, B., **Byler, N.**, Sutherland, R. S., Nicholls, D., Leitherer, C., & Stanway, E. R. (2019). “Comparison of Theoretical Starburst Photoionization Models for Optical Diagnostics,” *ApJ*, 878, 2. [ADS](#).
10. James, B., et al. including **N. Byler** (2019). “ASTRO2020 Science White Paper: Spatially Resolved UV Nebular Diagnostics in Star-Forming Galaxies,” *BAAS*, 51, 199. [ADS](#).
9. Aguado, D. S., et al. including **N. Byler** (2019). “The Fifteenth Data Release of the Sloan Digital Sky Surveys: First Release of MaNGA-derived Quantities, Data Visualization Tools, and Stellar Library,” *ApJS*, 240, 23. [ADS](#).
8. Leitherer, C., **Byler, N.**, Lee, J. C., & Levesque, E. M. (2018). “Physical Properties of II Zw 40’s Super Star Cluster and Nebula: New Insights and Puzzles from UV Spectroscopy,” *ApJ*, 865, 55. [ADS](#).
7. **Byler, N.**, Dalcanton, J. J., Conroy, C., Johnson, B. D., Levesque, E. M., & Berg, D. A. (2018). “Stellar and Nebular Diagnostics in the Ultraviolet for Star-forming Galaxies,” *ApJ*, 863, 14. [ADS](#).
6. **Byler, N.**, Dalcanton, J. J., Conroy, C., & Johnson, B. D. (2017). “Nebular Continuum and Line Emission in Stellar Population Synthesis Models,” *ApJ*, 840, 44. [ADS](#).
5. Choi, J., Conroy, C., & **Byler, N.** (2017). “The Evolution and Properties of Rotating Massive Star Populations,” *ApJ*, 838, 159. [ADS](#).
4. Leja, J., Johnson, B. D., Conroy, C., van Dokkum, P. G., & **Byler, N.** (2017). “Deriving Physical Properties from Broad-band Photometry with Prospector: Description of the Model and a Demonstration of its Accuracy Using 129 Galaxies in the Local Universe,” *ApJ*, 837, 170. [ADS](#).
3. Drory, N., et al. including **N. Byler** (2015). “The MaNGA Integral Field Unit Fiber Feed System for the Sloan 2.5 m Telescope,” *AJ*, 149, 77. [ADS](#).

2. Bundy, K., et al. *including N. Byler* (2015). "Overview of the SDSS-IV MaNGA Survey: Mapping nearby Galaxies at Apache Point Observatory," *ApJ*, 798, 7. [ADS](#).
1. Williams, B. F., et al. *including N. Byler* (2014). "The Panchromatic Hubble Andromeda Treasury. X. Ultraviolet to Infrared Photometry of 117 Million Equidistant Stars," *ApJS*, 215, 9. [ADS](#).